Advanced Wireless Communications Laboratory Series

Complexity-Adaptive LDPC-Coded OFDM under Fading and Interference

GNU Radio and Python simulation laboratory

Reproducible code, recorded measurements, and analysis

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Executive summary

This study compares fixed-iteration and syndrome-terminated normalized min-sum LDPC decoding in a QPSK-OFDM link subject to frequency-selective Rayleigh fading, additive noise, and narrowband interference. Each decoder is assessed on the same received soft information. The recorded experiment comprises 26,000 frames across 26 operating points and reports information-bit error rate (BER), frame-error rate (FER), and decoder iteration counts.

At code rate 1/2, SNR = 8 dB, and jammer-to-signal ratio (JSR) = −10 dB, 1,000 frames gave FER = 0.220 for fixed decoding and 0.216 at the first syndrome-valid decision. That decision occurred after 8.35 iterations on average, compared with a budget of 20. The first valid decision therefore required 58.25% fewer decoding iterations than the fixed budget.

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1 Introduction

LDPC decoding trades iterative processing for error-correction capability [1, 2]. Roth et al. implemented min-sum decoding with an early-termination procedure to avoid redundant iterations [3]. More recent work by Ren et al. addresses the reliability and implementation-cost trade-off through generalized adjusted min-sum decoding [4]. These studies motivate examining decoder effort alongside the error rate; neither algorithm is reproduced here.

This experiment compares the decision after a fixed iteration budget with the first decision satisfying all parity checks, using identical received frames. A subsequent iteration need not preserve an earlier valid decision. The comparison therefore measures both decoding reliability and the first valid iteration under fading, noise, and narrowband interference. The comparison uses the same LDPC code and decoder parameters for both decisions.

2 System architecture

The transmitter generates information bits, encodes them with a sparse systematic LDPC code, maps the codeword to QPSK, and constructs OFDM symbols. A six-tap fading channel, additive noise, and a single in-band tone produce the received waveform. After CP removal, FFT, one-tap equalization, and soft demapping, the decoder comparison uses a common LLR vector.

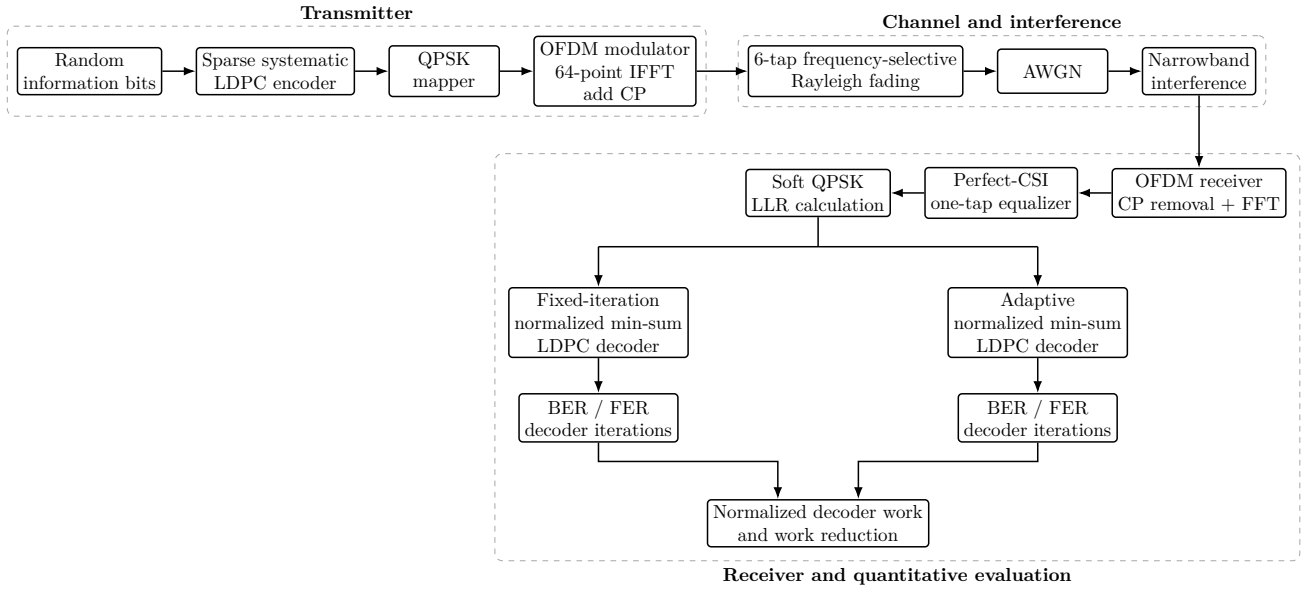


Figure 1: Complete LDPC-coded OFDM communication and evaluation architecture.

3 Signal-processing implementation

3.1 LDPC code

For each frame $K = 96$ information bits are encoded at rate $R \in \{1/2, 2/3, 3/4\}$. With $M = K(1-R)/R$ parity bits, the codeword length is $N = K + M$.

The parity-check matrix has the form

$$\mathbf{H} = [\,\mathbf{A}\ \mathbf{B}\,],$$

Table 1: Dimensions of the three constructed systematic codes.

R	Information bits K	Parity bits M	Codeword bits N
1/2	96	96	192
2/3	96	48	144
3/4	96	32	128

where each column of the $M \times K$ sparse matrix $\mathbf{A}$ contains three randomly placed ones and $\mathbf{B}$ is an $M \times M$ lower bidiagonal matrix with unit diagonal and subdiagonal. The transmitted codeword $\mathbf{c}$ satisfies $\mathbf{H}\mathbf{c}^T = \mathbf{0}$ over GF(2). The matrix construction is fixed by the simulation seed for each rate. These laboratory codes are not standardized 5G NR codes.

3.2 QPSK-OFDM waveform

Two coded bits map to

$$x = \frac{(1-2b_0) + j(1-2b_1)}{\sqrt{2}}.$$

The transmitter uses a 64-point IFFT, 48 occupied bins $\{1,\dots,24\} \cup \{40,\dots,63\}$, and a 16-sample cyclic prefix. For an OFDM frequency-domain vector $X[k]$,

$$x_{\mathrm{OFDM}}[n] = \frac{1}{\sqrt{64}} \sum_{k=0}^{63} X[k] e^{j2\pi kn/64}.$$

The codeword occupies two OFDM data symbols at every tested code rate; unused data-carrier positions in the final symbol are zero padded. The sample-rate setting is 64 kHz, giving 1 kHz nominal subcarrier spacing.

3.3 Fading, noise, and interference

A new quasi-static six-tap channel is generated for every coded frame[5]:

$$y_f[n] = \sum_{\ell=0}^{5} h[\ell] x[n-\ell], \qquad \mathbb{E}|h[\ell]|^2 = \frac{0.65^\ell}{\sum_{q=0}^{5} 0.65^q}.$$

The output is truncated to the frame length. Noise and a deterministic tone are added to form

$$y[n] = y_f[n] + w[n] + \sqrt{P_J}\, e^{j2\pi(12)n/64}.$$

With P_S denoting the measured mean power of the faded frame, the complex noise variance and interferer power are set by

$$P_N = P_S\, 10^{-\mathrm{SNR}/10}, \qquad P_J = P_S\, 10^{\mathrm{JSR}/10},$$

where SNR and JSR are specified in dB. The tone is centered on active FFT bin 12 (+12 kHz). This is an interference-impaired link; it does not estimate or cancel the interfering signal.

3.4 Equalization and soft information

The recorded study uses exact channel taps for one-tap frequency-domain equalization, $Z[k] = Y[k]/H[k]$. Frame timing and carrier synchronization are known. The complex Gaussian-noise variance, adjusted by the squared channel gain, is used in the approximate QPSK log-likelihood ratios:

$$L_I = \frac{2\sqrt{2}\,\mathrm{Re}\, Z[k]}{\sigma^2_{\mathrm{eff},k}}, \qquad L_Q = \frac{2\sqrt{2}\,\mathrm{Im}\, Z[k]}{\sigma^2_{\mathrm{eff},k}}.$$

The deterministic interferer is not included in the LLR variance model.

4 Decoder comparison

Both algorithms use normalized min-sum with $\alpha = 0.80$. This flooding-schedule decoder differs from the layered offset-min-sum hardware implementation in [3] and the generalized adjusted min-sum algorithm in [4]. For check node i and variable node j, the check-to-variable update is

$$r_{i\to j} = \alpha \left(\prod_{j' \in \mathcal{N}(i)\setminus j} \operatorname{sgn}(q_{j'\to i}) \right) \min_{j' \in \mathcal{N}(i)\setminus j} |q_{j'\to i}|.$$

After each complete check-node and variable-node update, the hard decision $\hat{\mathbf{c}}$ is examined using

$$\mathbf{s} = \mathbf{H}\hat{\mathbf{c}}^T \pmod 2.$$

The fixed reference returns its decision at $I_{\max}$ iterations. The adaptive decision is the first decision with $\mathbf{s} = \mathbf{0}$, or the decision at $I_{\max}$ if none is valid. Zero syndrome establishes codeword validity, not necessarily recovery of the transmitted information. For every received frame, the experiment records the first syndrome-valid decision and the decision after the fixed iteration budget. Their bit errors and iteration counts are evaluated using the same received soft information.

5 Experiment and measurement protocol

The experiment uses seed 2026 and 1,000 frames per operating point. The SNR and JSR sweeps, code-rate comparison, and iteration-limit comparison contain 26,000 frames in total.

Table 2: Recorded operating points. Each row represents a series of distinct 1,000-frame conditions.

Experiment	Varied parameter	Other settings	Frames
Representative	SNR = 8 dB	JSR = −10 dB, $R = 1/2$, $I_{\max} = 20$	1,000
SNR sweep	$0, 2, \ldots, 16$ dB	JSR = −10 dB, $R = 1/2$, $I_{\max} = 20$	9,000
JSR sweep	$-30, -25, \ldots, 0$ dB	SNR = 8 dB, $R = 1/2$, $I_{\max} = 20$	7,000
Rate comparison	$1/2, 2/3, 3/4$	SNR = 6 dB, JSR = −15 dB, $I_{\max} = 20$	3,000
Iteration budget	$5, 10, \ldots, 30$	SNR = 6 dB, JSR = −15 dB, $R = 1/2$	6,000
Total	26 conditions		26,000

The BER counts erroneous information bits, and the FER counts frames with at least one erroneous information bit:

$$\mathrm{BER} = \frac{\sum_f N_{e,f}}{96 N_f}, \qquad \mathrm{FER} = \frac{\sum_f \mathbf{1}\{N_{e,f} > 0\}}{N_f}.$$

For decoder iteration counts I_f, the iteration-based work proxy is

$$W = \frac{M}{96 N_f} \sum_{f=1}^{N_f} I_f, \qquad \eta_I = 100 \left(1 - \frac{W_{\text{adaptive}}}{W_{\text{fixed}}}\right) \%.$$

The work proxy is proportional to the number of decoder iterations. At a fixed code rate, its percentage reduction equals the reduction in mean iterations.

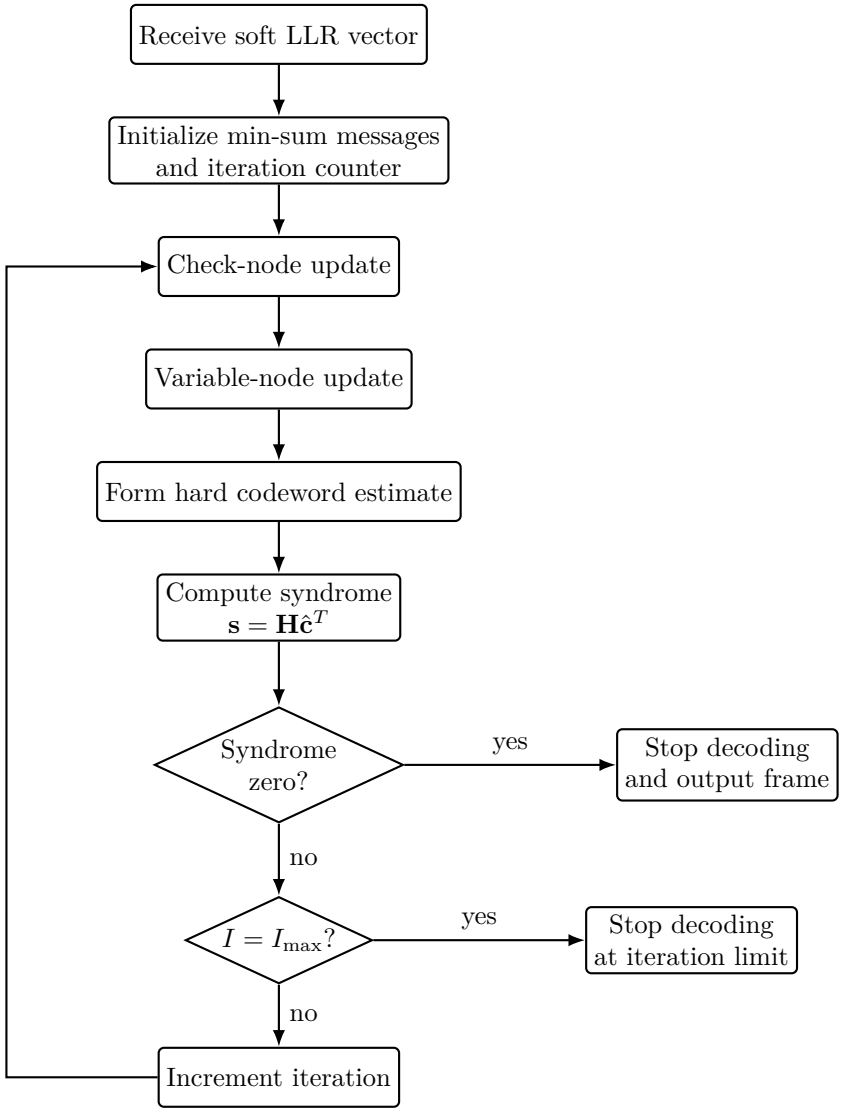


Figure 2: Syndrome-based adaptive LDPC early-termination logic

6 Recorded results

6.1 Representative received signals

Figure 3 shows the first 6,000 equalized QPSK values from the representative operating point. The four clusters remain identifiable, but their spread and outliers indicate substantial received impairment. The single-tone interferer is visible at +12 kHz in the normalized spectrum in Figure 3; that spectrum is not calibrated in dBm.

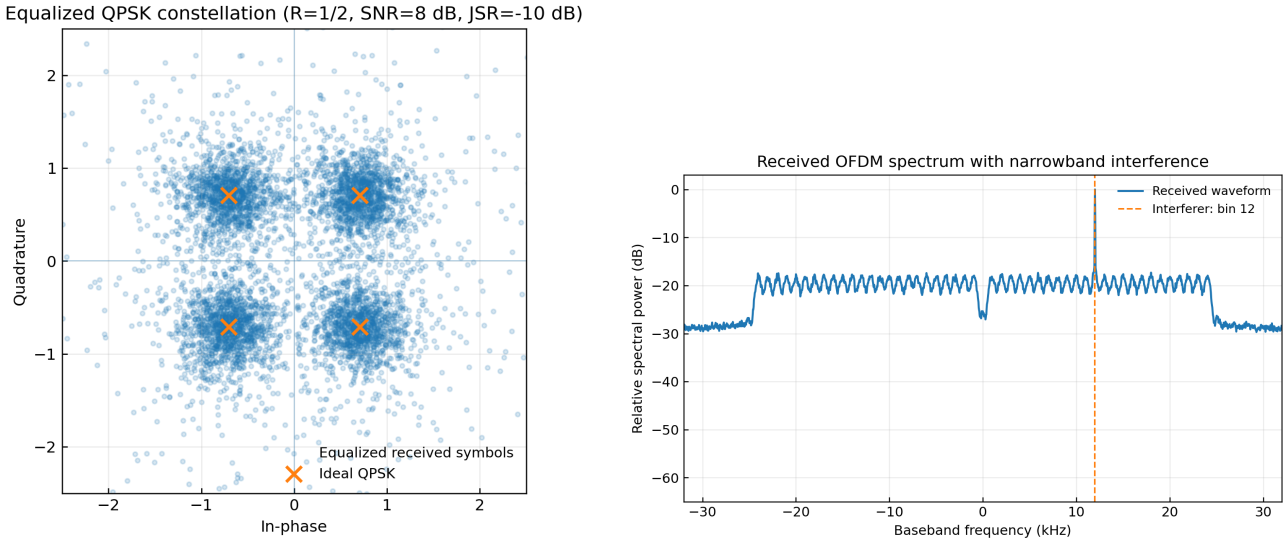


Figure 3: Representative received signals ($R = 1/2$, SNR $= 8$ dB, JSR $= -10$ dB): (a) equalized QPSK symbols and ideal decisions; (b) normalized received spectrum with an in-band tone on FFT bin 12.

The spectrogram in Figure 4 uses the recorded 160,000-sample receive buffer at 64 kHz. The horizontal axis is in milliseconds; the 2.5-second span represents concatenated simulated frames, not a continuous RF recording.

6.2 Representative decoding result

The fixed and first-valid decisions have FER 0.220 and 0.216, respectively. The first valid decision occurs after 8.35 iterations on average. This is 58.25% fewer iterations than the fixed limit of 20.

6.3 Noise sensitivity

Figures 6a and 6b use the nine SNR conditions at JSR $= -10$ dB. BER and FER decline with SNR while the recorded first-valid iteration moves earlier in the decoding budget. Over the SNR sweep, the iteration reduction increases from 1.97% at 0 dB to 68.28% at 16 dB. The deterministic interferer remains present at every point.

6.4 Interference sensitivity

At SNR $= 8$ dB, increasing JSR from -30 to 0 dB raises the first-valid FER from 0.017 to 0.763. The estimated iteration reduction falls from 86.49% to 14.33%. Strong interference produces more unsuccessful frames and forces the adaptive stopping condition to be reached later, or not at all before $I_{\max}$.

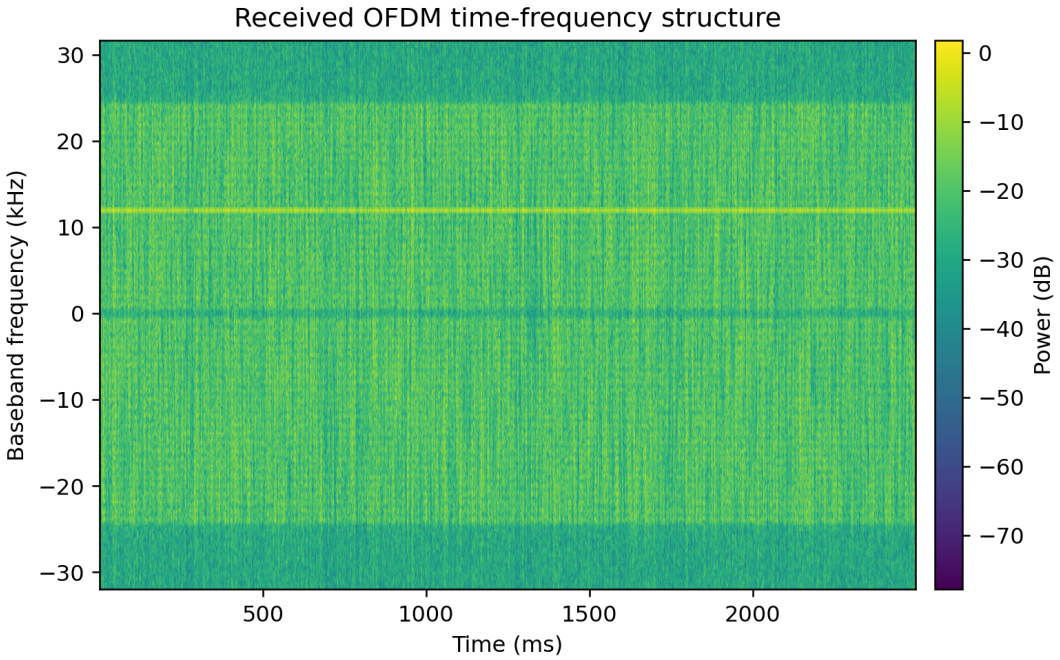


Figure 4: Time-frequency view of the representative received waveform. The persistent component at +12 kHz is the injected narrowband interferer.

Representative case | R=1/2, SNR=8 dB, JSR=-10 dB

Metric	Value
Fixed BER	0.0060729
First-valid BER	0.0060313
Fixed FER	0.22
First-valid FER	0.216
Fixed iteration budget	20
Mean first-valid iteration	8.35
Iteration reduction (%)	58.25
Frames	1000

Figure 5: Recorded decoding results for $R = 1/2$, SNR $= 8$ dB, JSR $= -10$ dB, and $I_{\max} = 20$, from 1,000 frames.

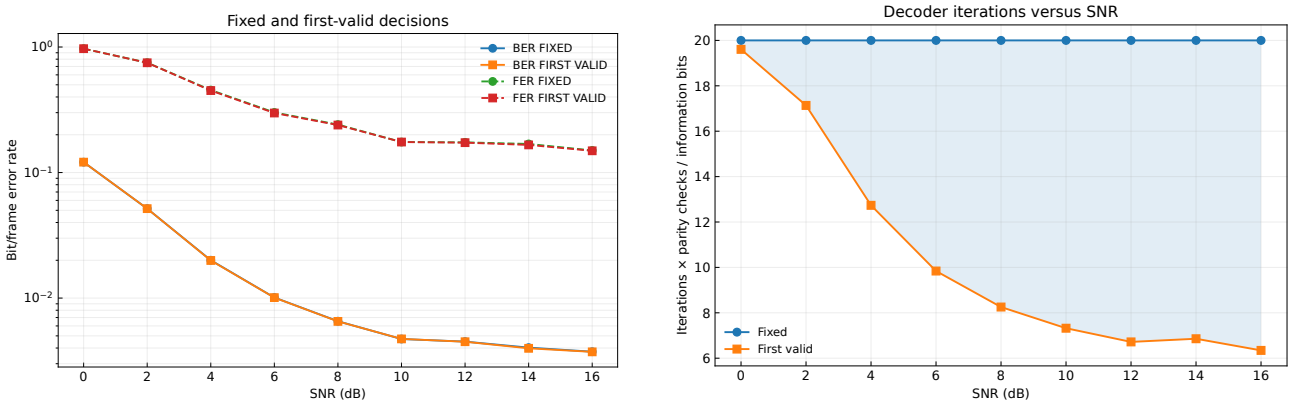


Figure 6: SNR sweep at $R = 1/2$, JSR $= -10$ dB: (a) measured BER and FER; (b) fixed iteration budget and estimated first-valid-decision work.

Table 3: Selected SNR-sweep values, distinct from the representative run.

SNR (dB)	Fixed BER	First-valid BER	Fixed FER	First-valid FER	Mean first-valid iterations
4	0.020021	0.019938	0.454	0.450	12.735
8	0.006542	0.006521	0.241	0.239	8.254
14	0.004031	0.003979	0.169	0.166	6.857

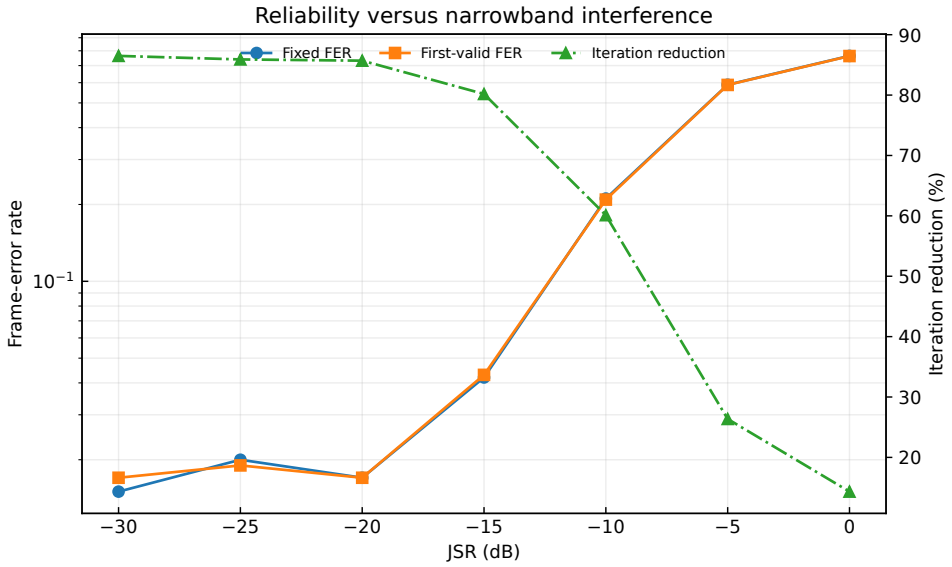


Figure 7: Measured FER and iteration reduction versus JSR at $R = 1/2$, SNR $= 8$ dB.

6.5 Rate and iteration comparisons

With SNR = 6 dB, JSR = −15 dB, and $I_{\max} = 20$, the measured first-valid FER increases from 0.093 at rate 1/2 to 0.235 at 2/3 and 0.374 at 3/4. Different code rates use independently generated parity-check matrices. This comparison does not isolate code rate from code structure.

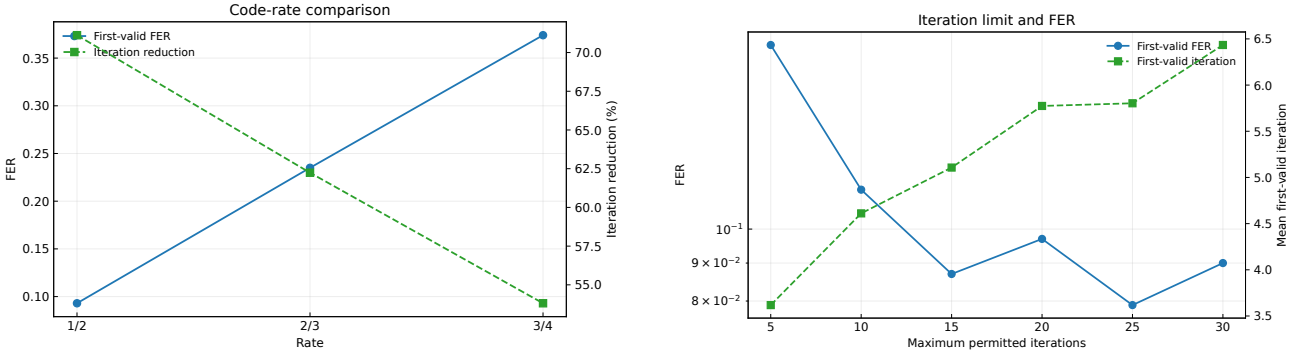


Figure 8: Rate and maximum-iteration experiments: (a) FER and iteration reduction for independently constructed codes; (b) FER and mean first-valid iterations at six different iteration budgets.

At $R = 1/2$, SNR = 6 dB, and JSR = −15 dB, increasing $I_{\max}$ from 5 to 30 changes first-valid FER from 0.177 to 0.090; the smallest observed FER, 0.079, occurs at a budget of 25 iterations. The six iteration-budget conditions use independently generated frames.

7 Implementation and reproducibility

The GNU Radio Companion flowgraph[6] contains an Embedded Python Block connected to constellation, frequency, waterfall, and numerical metric displays. It controls code rate, SNR, JSR, and decoder iteration budget.

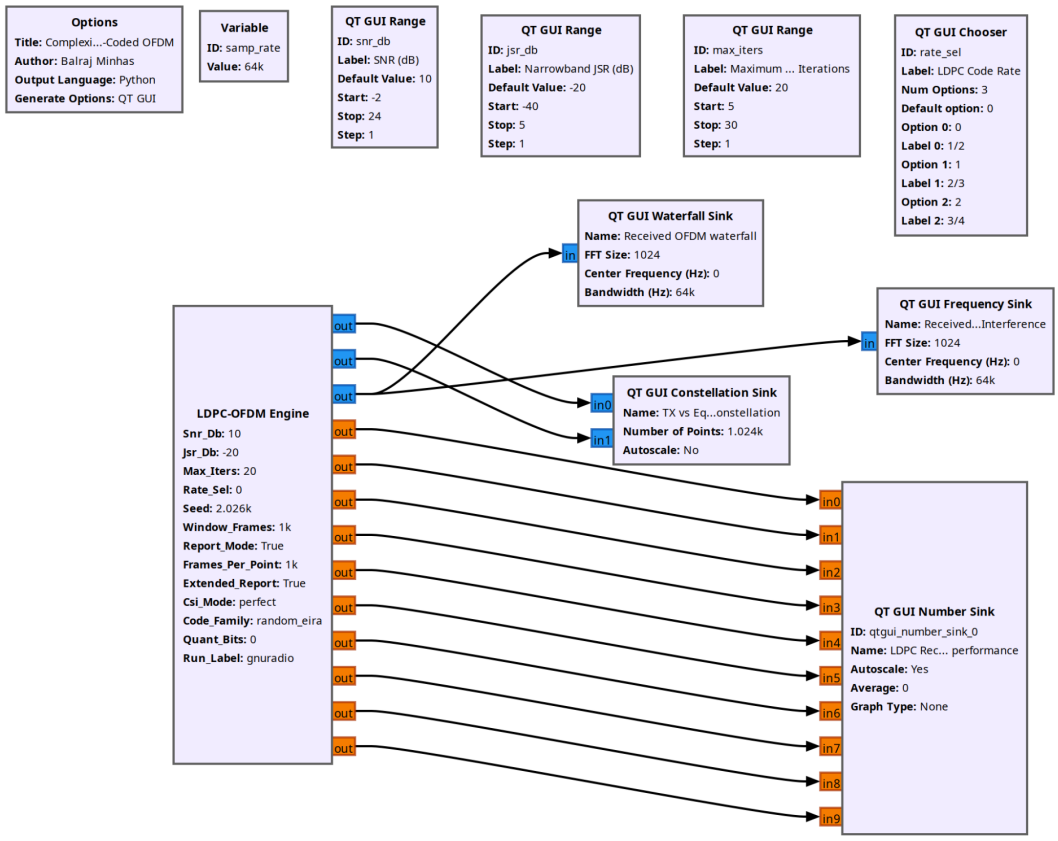


Figure 9: GNU Radio Companion flowgraph. The Embedded Python Block implements the coded-OFDM link and decoder comparison. QT GUI sinks display the received signals and decoder measurements.

The Python application and Embedded Python module are included with the flowgraph. The CSV contains one row per coded frame with error counts and decoder iteration measurements. The numerical results and figures use these records and the representative received-signal capture.

8 Scope and limitations

The experiment assumes perfect channel knowledge and known synchronization. The fading channel is constant within each frame and the interference is a single in-band tone. The LDPC code uses one parity-check matrix per rate. Each operating point includes 1,000 simulated frames.

9 Conclusion

The first syndrome-valid decision is often reached before the fixed decoding budget is exhausted. Iteration counts decrease as noise and narrowband interference weaken. The fixed and first-valid error rates remain close at the tested operating points. Syndrome-based stopping can therefore reduce the number of decoder message updates needed to obtain a valid decision.

References

[1] R. G. Gallager, "Low-density parity-check codes," *IRE Transactions on Information Theory*, vol. 8, no. 1, pp. 21–28, Jan. 1962.

[2] W. E. Ryan and S. Lin, *Channel Codes: Classical and Modern*. Cambridge University Press, 2009.

[3] C. Roth, P. Meinerzhagen, C. Studer, and A. Burg, "A 15.8 pJ/bit/iter quasi-cyclic LDPC decoder for IEEE 802.11n in 90 nm CMOS," in *Proc. IEEE Asian Solid-State Circuits Conf.*, 2010, pp. 1–4, doi: 10.1109/ASSCC.2010.5716618.

[4] Y. Ren, H. Harb, Y. Shen, A. Balatsoukas-Stimming, and A. Burg, "A generalized adjusted min-sum decoder for 5G LDPC codes: Algorithm and implementation," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 6, pp. 2911–2924, Jun. 2024, doi: 10.1109/TCSI.2024.3368056.

[5] D. Tse and P. Viswanath, *Fundamentals of Wireless Communication*. Cambridge University Press, 2005.

[6] GNU Radio Project, *GNU Radio Documentation*, `https://www.gnuradio.org/`.